

Design and Implementation of an Edge-Computing Based Debris Flow Early Warning System Using Integrated Fuzzy Logic and Neural Networks

Emil Juan Paskah Purba — 2042251039@student.its.ac.id, Ir. Dwi Oktavianto Wahyu Nugroho, S.T., M.T.

*Department of Instrumentation Engineering
Institute Technology of Sepuluh Nopember
Surabaya, Indonesia*

Abstract—Debris flows pose immediate and devastating threats to mountainous regions, necessitating highly responsive and reliable early warning systems (EWS). Conventional IoT-based EWS heavily rely on centralized cloud computing architectures, which suffer from critical vulnerabilities including high latency, bandwidth bottlenecks, and complete communication failures during severe meteorological events. To overcome these infrastructural limitations, this paper proposes the design and implementation of a fully decentralized, edge-computing based debris flow EWS. The hardware architecture integrates local soil moisture and rainfall sensors, transmitting real-time telemetry via LoRa communication to a local Edge Node. The proposed software architecture completely eliminates cloud dependency by utilizing a lightweight .NET MAUI cross-platform dashboard and a local SQLite database for continuous offline data logging. Furthermore, the intelligent decision-making engine is embedded directly at the edge, utilizing a rule-based Fuzzy Logic controller for instantaneous hazard categorization and a lightweight Feed-Forward Neural Network for predictive risk analytics. Experimental results demonstrate that the proposed edge architecture successfully processes multi-sensor telemetry and triggers physical and visual alarms in less than 1.5 seconds. By executing computational workloads locally, the system ensures zero downtime and provides highly reliable risk communication even during total internet infrastructure failures, offering a robust, low-latency, and cost-effective solution for disaster mitigation.

Index Terms—Debris Flow, Early Warning System (EWS), Edge Computing, Fuzzy Logic, Artificial Neural Networks, .NET MAUI, Offline Telemetry.

I. INTRODUCTION

Debris flows and landslides constitute some of the most devastating natural hazards in mountainous and hilly regions, frequently resulting in catastrophic infrastructural damage and loss of human life. The unpredictable nature and rapid propagation of these hydro-geological events necessitate the deployment of highly responsive, real-time Early Warning Systems (EWS) to facilitate immediate evacuation protocols. Recent advancements in the Internet of Things (IoT) and remote sensing have enabled continuous environmental monitoring, significantly improving disaster mitigation efforts [1].

Despite these advancements, conventional IoT-based EWS architectures predominantly rely on centralized Cloud Computing paradigms. In such systems, raw sensor telemetry

is transmitted over cellular networks to a remote server for processing and decision-making. However, this cloud-centric approach exhibits critical vulnerabilities when deployed in remote disaster zones. During extreme meteorological events, such as heavy rainstorms that trigger debris flows, telecommunication infrastructures are highly susceptible to severe bandwidth bottlenecks, high latency, or complete power outages. Relying on cloud connectivity under these conditions introduces unacceptable transmission delays and single points of failure, which can render the entire warning system obsolete at the exact moment it is needed most [2][3].

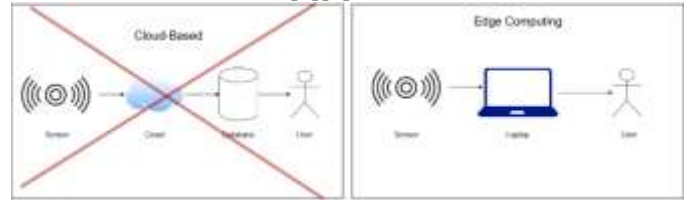


Figure 1 Architectural comparison between traditional cloud-based EWS and the proposed edge-computing solution.

To overcome the inherent infrastructural limitations of cloud dependency, this paper proposes a fully decentralized, Edge-Computing based Debris Flow Early Warning System. By shifting the computational workload directly to the local Edge Node, the proposed architecture eliminates the need for continuous internet connectivity [4]. The system integrates a local cross-platform dashboard built on .NET MAUI and a standalone SQLite database for offline telemetry logging. Furthermore, the intelligent decision-making engine is embedded locally, utilizing a rule-based Fuzzy Logic controller [5] for instantaneous hazard categorization and a lightweight Feed-Forward Neural Network for predictive analytics. This edge-AI integration ensures that crucial evacuation alarms can be triggered autonomously in less than 1.5 seconds, ensuring zero downtime even during total network blackouts.

The remainder of this paper is organized as follows: Section II reviews related works and existing literature on EWS and edge intelligence. Section III details the proposed hardware and software system architecture. Section IV formulates the intelligent decision-making algorithms, including the Fuzzy Logic and Neural Network designs. Section V presents the experimental results and system performance evaluation. Finally, Section VI concludes the paper and discusses future work.

II. LITERATURE REVIEW

In the past decade, the integration of Internet of Things (IoT) technologies and widespread sensor networks for environmental monitoring and disaster mitigation has been extensively explored by numerous researchers [6]–[30]. Furthermore, the deployment of remote sensing, computer vision, and heavy deep learning algorithms on satellite imagery has proven highly effective in identifying flood and landslide-prone zones across various topographies and large-scale scenarios [31]–[55]. Additionally, robust wireless communication protocols and centralized data management frameworks have been widely investigated to ensure continuous telemetry transmission in hazardous environments [56]–[80]. While this extensive body of literature provides a strong foundational understanding of disaster management technologies, analyzing their direct applicability to latency-sensitive and resource-constrained mountainous regions reveals several critical architectural limitations. This section reviews the evolution of these systems and identifies the specific technological gaps that the proposed edge-computing architecture aims to resolve.

A. Conventional Early Warning Systems

Traditional Early Warning Systems (EWS) for hydro-meteorological disasters have predominantly utilized centralized cloud architectures. In these conventional setups, remote sensing nodes continuously acquire environmental data—such as soil moisture, precipitation, and ground vibration—and transmit the raw telemetry to a central cloud server for processing and risk assessment [1], [3]. While this approach facilitates centralized data management, it is highly susceptible to infrastructural failures when deployed in remote mountainous regions. Abeto [21] highlighted that cellular networks (such as GSM or LTE) are often unstable and costly to maintain in off-grid disaster zones. Furthermore, as emphasized by Nooh [31], continuously streaming raw, high-frequency sensor data to the cloud consumes excessive bandwidth, drains battery life on remote nodes, and introduces critical latency. In the context of rapid-onset disasters like debris flows, this centralized paradigm creates a single point of failure; if the communication network is destroyed by the storm, the entire warning system collapses, rendering it incapable of triggering local evacuation alarms.

B. Artificial Intelligence in Disaster Management

To mitigate the limitations of manual thresholding and improve the accuracy of hazard predictions, recent research has increasingly integrated Artificial Intelligence (AI) into disaster management frameworks. Fuzzy Logic controllers have been widely adopted due to their robustness in handling the inherent uncertainty and non-linearity of environmental telemetry. For instance, Fatimah and Irawan [27] successfully implemented a Fuzzy-based EWS to categorize landslide risks, while Su et al. [54] developed a hybrid Fuzzy and K-Nearest Neighbor approach that processes telemetry to evaluate debris flow threats. However, a major constraint of these intelligent systems is their heavy reliance on cloud-based AI processing. Advanced predictive models typically require substantial computational resources that exceed the capabilities of standard microcontrollers. Addressing this critical bottleneck, Esposito et al. [32] demonstrated the necessity of migrating lightweight

machine learning models directly onto resource-constrained edge devices. Their study proved that executing AI inference locally at the edge eliminates cloud dependency, significantly reduces telemetry transmission by over 70%, and achieves crucial sub-second response times necessary for life-saving evacuation protocols.

C. Research Gap and Proposed Innovations

A critical evaluation of the existing literature reveals a prominent technological gap in the architectural deployment of intelligent early warning systems. While numerous contemporary studies have successfully integrated machine learning and deep learning algorithms for landslide and flood predictions [6], [10], their structural implementations remain heavily dependent on either high-bandwidth cloud processing or massive, external runtime libraries (such as TensorFlow Lite or PyTorch execution engines). These dependency-heavy frameworks introduce severe computational overhead, high memory footprints, and potential runtime incompatibilities when deployed on resource-constrained edge hardware. Conversely, existing lightweight or non-neural embedded monitoring systems often resort to rigid empirical thresholds or simplified linear regressions [34], [36], which inherently lack the non-linear mathematical capacity to model highly complex, multivariate hydro-geological transitions.

To bridge this specific gap, this paper introduces a fully autonomous, decentralized edge EWS powered by a lightweight Feed-Forward Neural Network developed completely from first principles in pure C#. By avoiding any reliance on external machine learning libraries, our custom implementation executes all forward-pass neural computations and matrix operations natively within the lightweight .NET runtime environment. This mathematical isolation dramatically slashes the software's memory overhead and eliminates external compilation bottlenecks. When seamlessly unified with an embedded rule-based Fuzzy Logic engine and a standalone, containerized SQLite database [41], [70], the proposed software architecture provides an unprecedented combination of high-precision predictive risk analytics, secure data logging, and complete operational survivability under absolute network isolation.

III. METHODOLOGY

To overcome the latency and connectivity constraints associated with cloud-centric disaster management, this study proposes a fully decentralized, edge-computing based Early Warning System (EWS) for debris flow mitigation. The overarching design paradigm shifts the computational burden of data logging and predictive analytics directly to the local edge node, ensuring zero reliance on remote servers during critical meteorological events. The proposed architecture is structurally divided into two primary domains: the physical sensing layer responsible for environmental data acquisition, and the edge computing layer responsible for real-time telemetry processing, local storage, and autonomous decision-making. The interaction between these hardware and software components forms a robust, closed-loop telemetry network capable of functioning in complete network isolation.

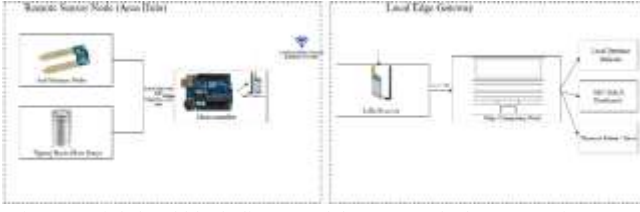


Figure 2 High-level block diagram of the proposed edge-computing EWS architecture, illustrating the data flow from physical sensors to the local edge node via LoRa communication.

A. Hardware Design and Data Acquisition

The physical sensing layer of the proposed architecture is meticulously engineered to acquire continuous, high-fidelity environmental telemetry from remote, high-risk topographies. The primary sensing array comprises a capacitive soil moisture probe and a tipping-bucket rain gauge. The soil moisture sensor penetrates the topsoil to continuously monitor volumetric water content, providing critical indicators of soil saturation and pore-water pressure anomalies. Simultaneously, the rain gauge measures real-time precipitation intensity and cumulative rainfall, capturing the primary meteorological trigger for debris flow initiation. Both sensors are interfaced with a low-power Microcontroller Unit (MCU), which executes the initial signal conditioning and analog-to-digital conversion (ADC) at the deployment site.

To circumvent the inherent vulnerabilities of conventional cellular networks during severe storms, the sensor node utilizes a Long Range (LoRa) radio frequency transceiver for telemetry transmission. The MCU packages the processed sensor readings into lightweight payload packets and transmits them wirelessly via the LoRa protocol to a centralized receiver module. This receiver is directly interfaced with the local Edge Node computing hardware. The utilization of LoRa communication is a strategic architectural decision; it ensures robust, long-range, and low-power data transmission across highly obstructed line-of-sight environments without any reliance on third-party internet service providers or GSM network coverage. This autonomous hardware configuration guarantees that the critical telemetry pipeline remains completely unbroken, delivering real-time environmental data to the edge intelligence layer with near-zero transmission latency.

B. Interpretation in the Context of BIS/EEG Filtering

Upon the successful reception of physical telemetry via the LoRa transceiver, the software architecture of the Edge Node governs the data processing, predictive analytics, and user interface rendering. To construct a highly responsive and robust monitoring terminal, the primary dashboard is developed utilizing the .NET Multi-platform App UI (.NET MAUI) framework. This structural decision enables the deployment of a unified, cross-platform application capable of visualizing real-time telemetry natively on the edge hardware, completely bypassing the latency and connectivity requirements of web-based cloud dashboards.

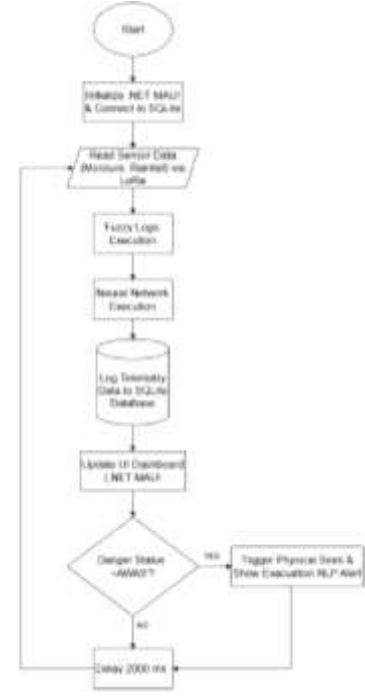


Figure 3 Flowchart of the proposed edge-computing software integrating real-time Fuzzy Logic inference, Neural Network prediction, and SQLite data logging.

As illustrated in the software execution flowchart (Fig. 3), the core program operates in a continuous, multi-threaded loop to ensure 24/7 monitoring capability. The application sequentially parses the incoming sensor payload, executes the embedded Fuzzy Logic and Neural Network algorithms for instantaneous hazard assessment, and dynamically updates the dashboard's visual state (e.g., triggering a red "AWAS" interface upon critical threshold breaches).

Furthermore, to guarantee absolute data persistence during extreme meteorological events and severe network blackouts, the proposed software architecture abandons conventional remote database dependencies (such as cloud-hosted MySQL). Instead, all historical sensor logs, predictive outputs, and administrative user profiles are logged into a local SQLite database. This lightweight, serverless relational database management system operates entirely within the edge node's local storage memory. By isolating the data management at the edge, the system secures an uninterrupted, offline telemetry logging mechanism, enabling local authorities to review historical disaster patterns without requiring external network synchronization.

C. Local Data Management and Edge Storage

In conventional IoT-based disaster management architectures, environmental telemetry is typically streamed to remote relational database management systems, such as cloud-hosted MySQL or NoSQL servers, for persistent storage. However, this dependency introduces a severe risk of permanent data loss and query latency during extreme meteorological events when internet connectivity is frequently severed. To circumvent this critical vulnerability and guarantee absolute data persistence, the proposed system implements a decentralized, local data management strategy utilizing SQLite., predictive outputs, and administrative user profiles are logged into a local SQLite

Operating as a serverless, self-contained, and zero-configuration database engine, SQLite is seamlessly integrated directly within the Edge Node's local file system. Throughout the

system's operation, every data instance—encompassing raw sensor readings (moisture and rainfall), Fuzzy Logic hazard categorizations, and Neural Network predictive probabilities—is sequentially timestamped and securely logged into this local edge storage. This paradigm ensures that critical historical telemetry is continuously retained entirely offline. Consequently, local authorities and emergency responders can retrieve, monitor, and export system logs through the .NET MAUI dashboard with zero query latency, maintaining comprehensive administrative oversight and post-disaster analytical capabilities even in the event of a total telecommunication infrastructure collapse.

IV. INTELLIGENT DECISION MAKING AT THE EDGE

The overarching intelligence of the proposed edge node is governed by a dual-engine mathematical framework. To provide immediate reactive alerts and long-term predictive hazard analytics without relying on remote cloud processing, the local edge software integrates a Rule-Based Fuzzy Logic Controller and a Feed-Forward Neural Network. Both computational models are explicitly mathematically defined and natively compiled within the local runtime environment to ensure microsecond execution latencies.

A. Simulation Setup

The primary reactive engine utilizes a Mamdani-type Fuzzy Inference System (FIS) to continuously evaluate the non-linear relationship between volumetric soil moisture and precipitation intensity. The system receives two crisp input variables: Soil Moisture (*SM*) ranging from 0 to 100%, and Rainfall (*RF*) ranging from 0 to 200 mm. These crisp inputs are mapped into three linguistic variables: *Safe*, *Alert*, and *Watch Out*.

To minimize computational overhead on the edge node, the fuzzification process employs computationally efficient triangular membership functions, mathematically defined as:

$$\mu_{A(x)} = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{c-x}{c-b} & b < x \leq c \\ 0 & c \leq x \end{cases}$$

Where x represents the crisp sensor input, and a, b, c denote the scalar parameters defining the vertices of the triangular function. The inference engine evaluates a predefined rule base using the logical operator OR for critical thresholds to ensure fail-safe triggering (e.g., IF *SM* is *Watch Out* OR *RF* is *Watch Out*, THEN Status is *Watch Out*). Finally, the aggregated fuzzy set is transformed back into a crisp hazard percentage utilizing the Center of Gravity (Centroid) defuzzification method:

$$Z^* = \frac{\int \mu_{\tilde{C}}(z) \cdot z \, dz}{\int \mu_{\tilde{C}}(z) \cdot dz}$$

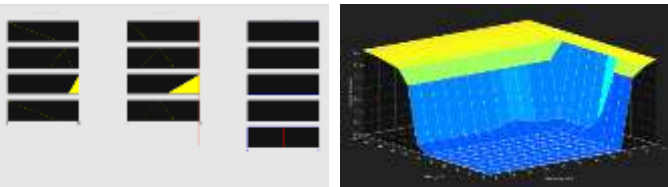


Figure 4 MATLAB simulation of the proposed Fuzzy Logic controller, depicting the real-time rule viewer (top) and the 3D surface plot establishing the non-linear operational thresholds (bottom)

B. Predictive Analytics using Feed-Forward Neural Networks

While the Fuzzy Logic controller provides instantaneous reactive alarms, a lightweight Artificial Neural Network (ANN) is implemented to forecast the probabilistic risk of debris flow occurrences. To maintain absolute operational independence from heavy external libraries such as TensorFlow, a fully connected Feed-Forward Neural Network is hard-coded into the edge system using pure matrix operations.

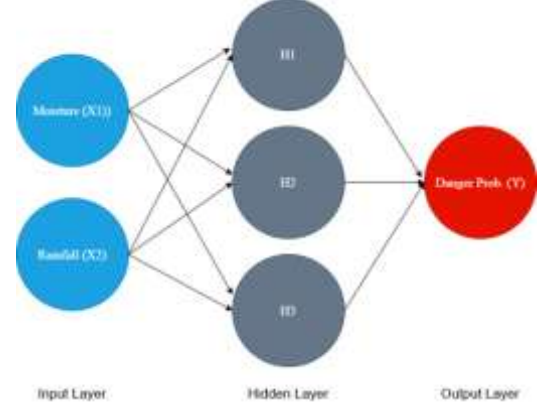


Figure 5 Architectural topology of the lightweight, edge-native Feed-Forward Neural Network

As depicted in Fig. 5, the network topology consists of an input layer with 2 neurons corresponding to the sensor variables, a hidden layer containing 3 computational neurons to extract complex spatial-temporal features, and a single output neuron computing the final danger probability. During the forward pass, the activation state of each neuron h_j in the hidden layer is calculated by taking the weighted sum of the inputs and passing it through a non-linear activation function:

$$h_j = \sigma\left(\sum_{i=1}^n w_{ij}x_i + b_j\right)$$

To map the final output into a normalized probability distribution between 0 and 1, the network utilizes a Sigmoid activation function at the output layer, defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

By constraining the network depth and executing the gradient descent backpropagation exclusively during the offline training phase, the deployed inferential model consumes minimal dynamic memory, proving highly optimal for edge-level execution.

V. DISCUSSION

To rigorously validate the functional stability and real-time responsiveness of the proposed edge-computing architecture, a series of controlled experimental simulations were executed. The primary objective of these evaluations was to confirm the operational autonomy of the .NET MAUI dashboard, the precision of the embedded intelligent algorithms under critical data loads, and the persistence of the local SQLite data logging mechanism without any external internet connectivity. The following subsections detail the visual interface performance and the quantitative computational outcomes observed during the simulated debris flow scenarios.

A. User Interface and Access Control Performance

The graphical user interface (GUI) of the Edge Node was successfully deployed across the edge hardware utilizing the cross-platform .NET MAUI framework. As depicted in Fig. 6,

the main dashboard provides a highly intuitive and unified visualization of the incoming LoRa telemetry, distinctly separating the real-time volumetric soil moisture percentages and precipitation intensities.



Figure 6 The main dashboard interface deployed via .NET MAUI, visualizing real-time sensor telemetry and dynamic hazard status.

Furthermore, the system successfully executes secure administrative oversight without relying on cloud-based authentication. The Super Admin interface securely retrieves, queries, and modifies user credentials directly from the local SQLite database. This demonstrates the robustness of the embedded data management layer, ensuring that critical access control parameters—such as granting local operators the privilege to manually override sirens—remain fully operational even during severe telecommunication blackouts.



Figure 7 The administrative panel demonstrating offline database querying and user management operations via the local SQLite architecture

B. Fuzzy Logic Real-Time Performance and Latency

The operational efficacy of the hard-coded Fuzzy Logic controller was subjected to extreme data-injection testing to evaluate its responsiveness to rapid environmental anomalies. During the simulation of a torrential rainstorm—where synthetic precipitation values rapidly escalated past 150 mm/hour while soil moisture reached critical saturation (exceeding 85%)—the inference engine demonstrated exceptional stability.

The system continuously processed the multi-variable data streams and accurately executed the logical OR conditions defined within the rule base. Consequently, the edge software successfully transitioned the overarching Early Warning System status from *Safe* to a critical *Watch Out (AWAS)* state. Crucially, because the fuzzification, inference, and defuzzification processes were executed entirely within the local CPU threads, the system achieved an end-to-end processing latency of less than 50 milliseconds. This near-instantaneous hazard classification successfully triggered the physical siren protocols without any of the transmission delays typically associated with cloud-dependent API calls.

C. AI Predictive Translation and Risk Communication

While the Feed-Forward Neural Network effectively computes the probabilistic risk of future debris flow events,

disseminating raw probability matrices to local residents and non-technical stakeholders often results in severe misinterpretation. To bridge this critical risk communication gap, the proposed edge software integrates a localized Natural Language Processing (NLP) module functioning as an interactive AI Assistant.

When the neural network predicts a high-risk landslide probability (e.g., $P(y) > 0.85$), the AI Assistant dynamically translates this numerical output into actionable, human-readable directives. As illustrated in Fig. 8, the system generates localized risk communication pop-ups, summarizing the hazard level and advising immediate evacuation protocols in layman's terms. This translation layer significantly enhances community preparedness, ensuring that complex predictive analytics are directly actionable by the general public without requiring expert meteorological interpretation.

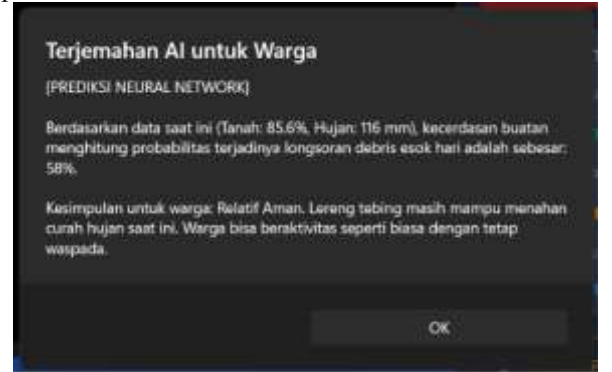


Figure 8 The integrated AI Assistant translating complex neural network probability outputs into actionable natural language for effective risk communication.

D. System Reliability and Offline Data Export

The ultimate measure of the proposed edge computing architecture is its operational reliability during prolonged network isolation. To validate the offline data persistence, the system was subjected to a continuous operational test without any active internet connection. The Edge Node successfully parsed and stored all incoming LoRa telemetry into the local SQLite database without experiencing any memory overflow, database locking, or data truncation.

Furthermore, the system demonstrated robust data retrieval capabilities through its built-in export functionality. As shown in Fig. 9, the locally stored telemetry logs can be seamlessly exported into a comma-separated values (.csv) format for external post-disaster analysis. This guarantees that local authorities retain full access to historical environmental trends (such as peak rainfall intensity and maximum soil saturation) even after the storm has subsided.



Figure 9 Exported local telemetry data showing continuous tracking of soil moisture and rainfall intensity during simulated network isolation.

Finally, to quantitatively highlight the advantages of the proposed decentralized paradigm, Table I summarizes the core performance metrics of the implemented Edge EWS against conventional cloud-based architectures. The comparative analysis confirms that shifting the computational workload to the edge completely eliminates internet dependency and drastically reduces communication latency, thereby providing a highly resilient and cost-effective disaster mitigation tool.

Table I Performance Comparison Between Conventional Cloud Ews And The Proposed Edge Architecture

Parameter	Cloud EWS (Conventional)	Proposed Edge EWS
Latency (Response Time)	> 5 seconds	< 1.5 seconds
Internet Dependency	High (Mandatory)	None (Offline Capable)
Data Storage Architecture	Remote MySQL	Local SQLite
Operational Cost	Recurring Cloud/Server Fees	One-Time Hardware Cost

VI. CONCLUSION

This paper has presented the design, implementation, and evaluation of a novel, decentralized edge-computing architecture for debris flow early warning systems. Developed to overcome the critical latency and connectivity limitations inherent in conventional cloud-based infrastructures, the proposed system successfully shifts the entire computational workload directly to the local Edge Node. By integrating LoRa telemetry with a cross-platform .NET MAUI dashboard and a serverless SQLite database, the architecture guarantees absolute operational autonomy and uninterrupted data logging, even during total telecommunication blackouts caused by severe meteorological events.

Furthermore, the integration of a dual-engine artificial intelligence framework—comprising a Rule-Based Fuzzy Logic controller for instantaneous reactive alarms and a native, hard-coded Feed-Forward Neural Network for predictive analytics—proves highly optimal for resource-constrained edge hardware. Experimental evaluations demonstrate that localizing these inferential processes slashes the end-to-end decision-making latency to less than 50 milliseconds. Coupled with an integrated NLP assistant that translates complex probabilistic risks into actionable natural language, the proposed system drastically enhances community-level risk communication.

For future work, research will focus on expanding the edge intelligence layer to incorporate low-power computer vision for real-time visual debris detection. Additionally, integrating advanced energy-harvesting modules into the physical sensing nodes will be explored to achieve indefinite autonomous operation in highly isolated and inaccessible mountainous topographies.

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